Agentic AI for Knowledge Production

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Level-setting and getting started

Ground rules

- The curse of knowledge is binding
- Be your most aggressive seminar selves
- This should be interactive (not a lecture)

Please make sure that you environment has been standardized!

Today's Agenda

Level-setting and getting started

What makes your chatbot "agentic"?

A shared vocabulary

Demo: data scraping (Sankalp)

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Take the maximally AI-hostile view

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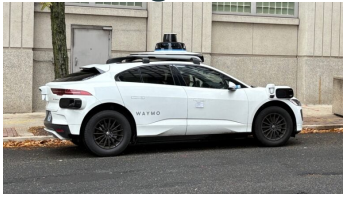
A computer cannot be held accountable.

Therefore a computer must never make a management _____

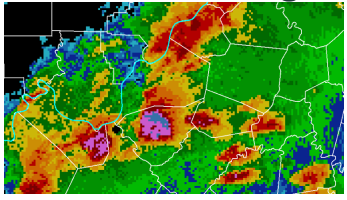
51722 7595 665 3779 413 7504 60757 13 19233 261 7595
2804 3779 1520 261 6411 _____

The predict-the-next-token framework generalizes

Self-driving cars



Weather forecasting



Chess

Note: The way you define a "token" carries some semantic meaning

Why should the AI skeptic care?

"Relational consequentialism"

- Fallen in love
- Been manipulated in strategic interactions
- Been convinced to commit murder/suicide
- Participate in economic exchange

"Deontological" grounding

- The bots have been trained to have the 3Hs: Helpful, Harmless, Honest
- Claude has a "Constitution"
- Grok revulsion
- Ideological alignment

Direct consequentialism: They can manipulate *your* world

A lucky coincidence?

- The next-token-prediction task for natural language happens to be close to the next-token-prediction task for formal (computer) language
- We built formal language to
  - Systematize information
  - Process it, *quickly*

Formal language tooling

Category	Examples / Description
No-code tools	Zapier, Tableau, Excel, Airtable
High-level languages	Python, R, JavaScript, Ruby
System Languages	C, C++, Rust, Go
Assembly Language	ASSEMBLY
Instruction set architecture	x86(-64), ARM, RISC–V
Electrical engineering	

A computational hierarchy: from no-code tools to hardware

Back to our timeline

Nov. 2022: ChatGPT

- We started producing a huge amount of training data in our conversations

> `Write me a sonnet about my windowless PhD carrell`

> `No, make it Petrarchan. No, Line 4 doesn't have the right meter. No, the rhyme scheme is off`

Back to our timeline

Mar. 2024: o1

- The first "reasoning model"

> `How many 'r's are there in 'strawberry'?`

Reinforcement learning

At a 30,000 ft view:

- You can offer a high-level objective
- And an *ex interim* reward function
- The bots begin steering *themselves*

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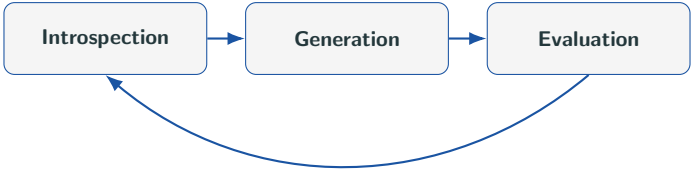
An OODA loop:

The bots can do



Where have we seen big performance uplifts?

Mathematics

- We have strong notions of theorems, definitions, contradictions, problem solving techniques
- To say nothing of formal proof assistance tools (LEAN) Evals

Software engineering

- The formal language problem was so hard that we developed a huge amount of tooling Evals

Arriving in the agentic era

Combine two observations:

1. Next-token-prediction can do an unreasonably good job at translating natural language intent ot formal structure
2. "Reasoning" with introspection improves output recursively in a way that makes computer-use reliable

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"We give the agents tools"

There is only *one* tool: the general purpose computer

The agent's tool happens to be the same as our tool

A shared vocabulary

Stochastic properties

- **Context:** Information that is project/repo/user/session-specific. Places soft bounds on the domain of knowledge and informs the agent's behavior
- `CLAUDE.md`**:** A specific instantiation of context that is ready by Claude at the beginning of every session. This soft-binds Claude's behavior (but NOT other agents). Can be defined at the user/project/repo/folder level.
- **Memory:** Discrete snippets of context generated and read by Claude that can be referred by Claude to infer context
- **(Sub-)Agents:** An instance of Claude. Can be invoked and dispatched explicitly or intelligently ith scoped tasks/context.
- **Skills:** A recipe detailing a particular procedure when invoked. Can be invoked explicitly or intelligently.
- **Rules:** A restricted version of context that is scoped narrowly to particular file types, directories, or tools

Deterministic properties

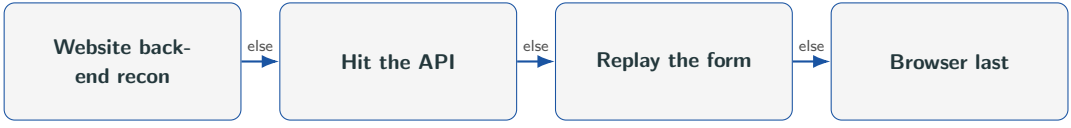
- **Hooks:** deterministic actions that are executed ONLY upon user-determined shell events. Cannot be invoked on within-CoT events.
- **Harness:** an environment (including system prompts, hooks, deterministic processes) that constrains and guides a model, *NOT an agent* in computer use
- **Worktrees:** A local clone of a repository to permit clock-simultaneous modifications of a single file, locally (by multiple agents)
- **Model Context Protocol (MCP):** A machine protocol (a subset of APIs) designed to inform an LLM of the properties and capabilities of the that protocol's host service.

Demo: data scraping (Sankalp)

Finding and collecting data

1. Be careful with government websites
2. Don't scrape what you don't needs
3. **ALWAYS** check robots.txt

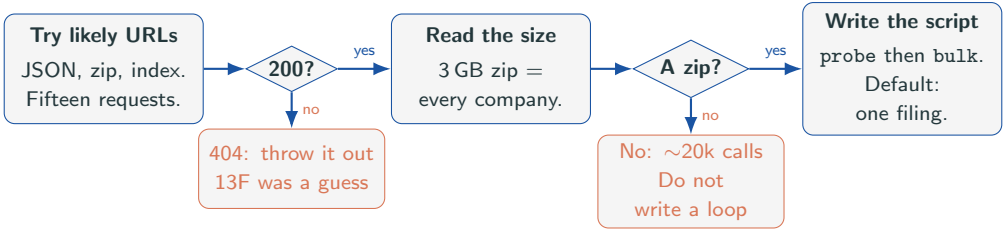
TODAY's example: SEC Edgar filings
Workflow:



Worked example: `fall-2026/sec_edgar.py`.

How we wrote `sec_edgar.py`

We did not scrape the search page. We asked what it talks to.



Every request needs a `User-Agent` with your name and email. Without it you get 403, not data.

On a new site, fingerprint the HTML first

Fetch the page (or have the agent fetch it) and search the source. The markers below pick the method.

Marker	First move
`.zip`, `.csv`, Download	Take the file.
`/api/`, `.json`	Call it. Skip the HTML.
`MapServer`, `FeatureServer`	ArcGIS. Query the layer; start with a count.
`__VIEWSTATE`, `.aspx`	WebForms. Replay the POST. Hidden tokens go stale after each click.
`_csrf`, `JSESSIONID`	Session token. Keep the cookie across the run.

Then hit the candidate URLs. Record status, size, row count, whether auth is real, and the pagination parameter (`page`, `from`, `offset`).

Write `scrape/sitename_endpoints.md`. Pick the extract after that file exists.

What you tell the agent

Skips the cheap path

```
Scrape this website and
give me a CSV.
```

Finds it

```
Probe candidate URLs.
Report status, size,
record count, and auth.
Write endpoints.md.
Do not write a scraper yet.
```

Once the path is known, from `sec_edgar.py`:

Habit	Why
Safe default	No arguments: one filing under 500 KB. Bulk needs `--yes`.
Write, then rename	A crash mid-write otherwise leaves a short file that resume treats as done.
Test URLs offline	Builders and parsers have no network. `selftest` is `assert`.
One bad record	At 20,000 requests some fail for reasons unrelated to your code. Crashing on item 4,000 is how you lose a night.

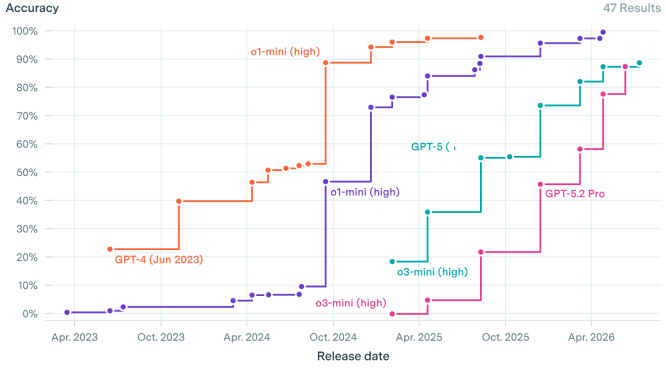
Starting prompt

Paste this first. Stop when the file exists.

```
Target: https://www.sec.gov/edgar/
Goal: find the real backend/API behind EDGAR and
document every bulk data path.
Deliverable: scrape/sec_edgar_endpoints.md -- host,
base paths, auth, record counts, pagination params,
bulk-download sizes, rate limits, with live
verification evidence.
Depth: recon only. Do not build a scraper yet.
```

LLM Evals: Mathematics

Frontier performance across benchmarks



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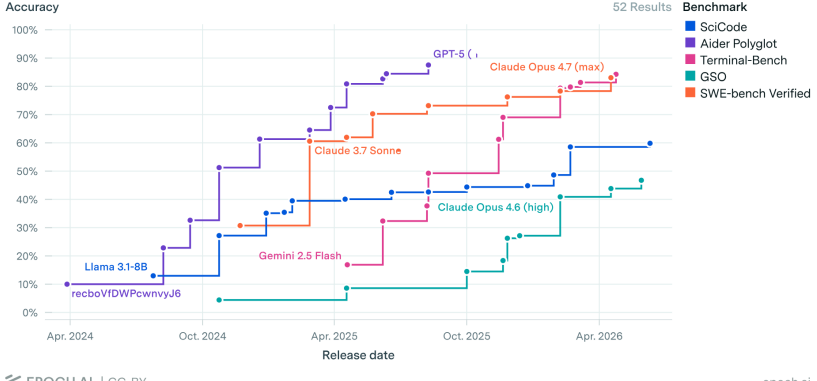
LLM Evals: Software Engineering

Frontier performance across benchmarks



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